MOVIE RECOMMENDATION SYSTEM

Subtitle: Jacaranda Movie Shop - Personalized Movie Suggestions

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PROJECT OVERVIEW

- **Business Goal:** Design a recommendation system to increase user engagement, satisfaction, and revenue for Jacaranda Movie Shop.
- **Data Source:** MovieLens Small Dataset
- **movies.csv:** Contains movie titles and genres.
- **ratings.csv:** Contains user ratings and timestamps.
- **tags.csv:** Contains user-generated tags.
- **links.csv:** Provides links to IMDb and TMDb.
- **Key Tasks:**
- Explore user preferences and genre trends.
- Build a machine learning model to recommend 5 movies based on a user's past ratings.

DATA EXPLORATION - USER & MOVIE STATISTICS

We analyzed the distribution of ratings to understand user and movie popularity.

User Activity (Top 20 users by rating count):

•Users can rate anywhere from 836 to 2698 movies, showing a wide range of engagement.

Movie Ratings (Top 20 movies by average rating):

•Movies with a 5.0 average rating (e.g., movieId 99) have very few ratings (often 1 or 2), making them statistical outliers.

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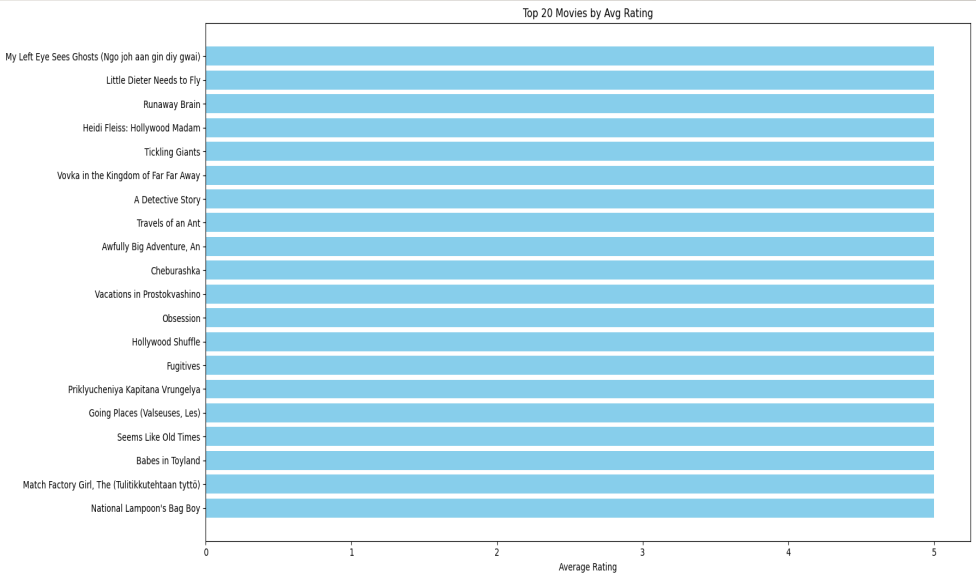
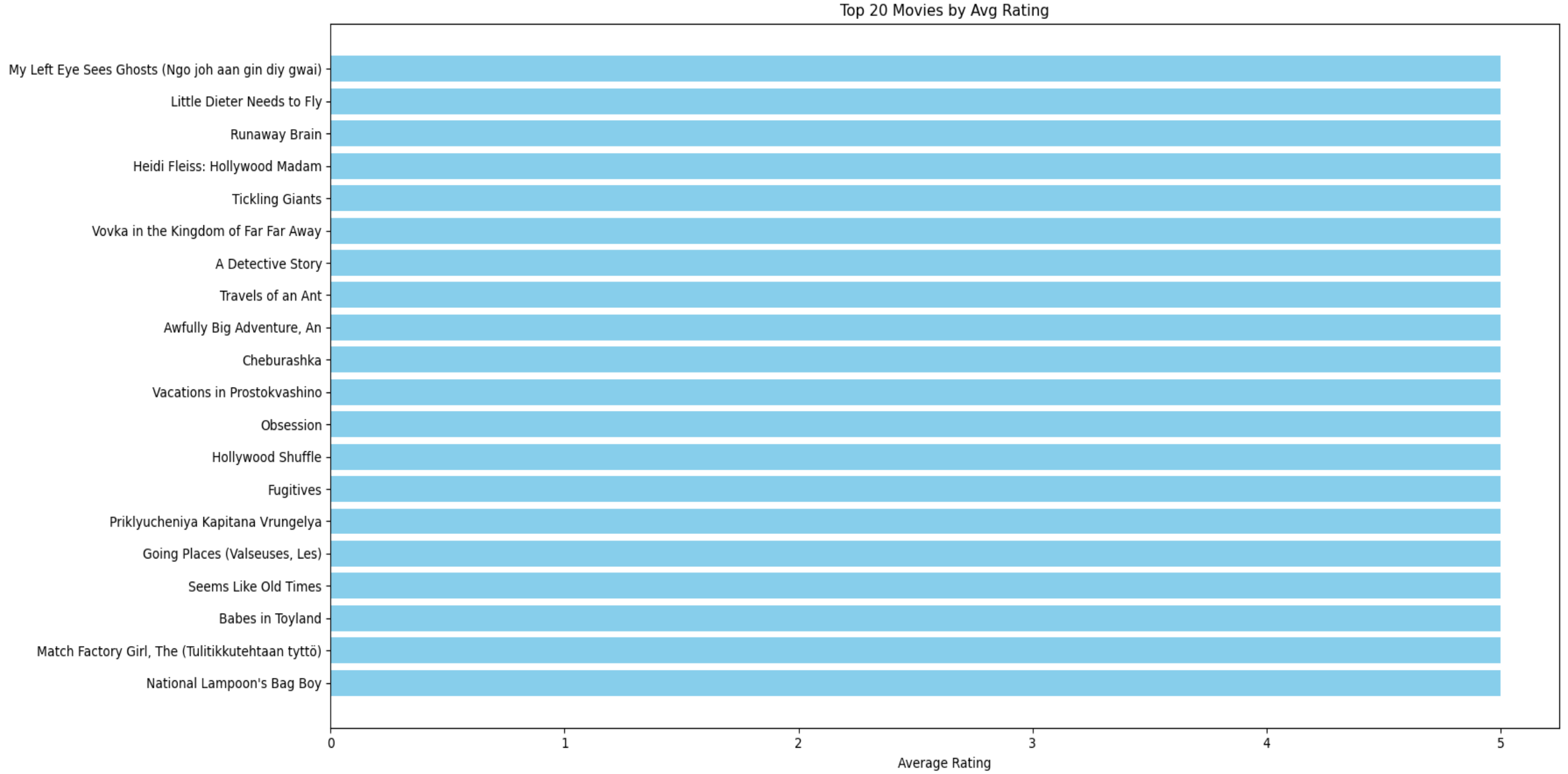
- **Movie Popularity (Top 20 movies by total rating count):**
- The most watched movies are popular classics like *Forrest Gump*, *The Shawshank Redemption*, and *Pulp Fiction*, each with over 300 ratings.
- **Interpretation:** High average ratings don't always mean popularity. A hybrid approach considering both rating count and average rating is needed for robust recommendations.

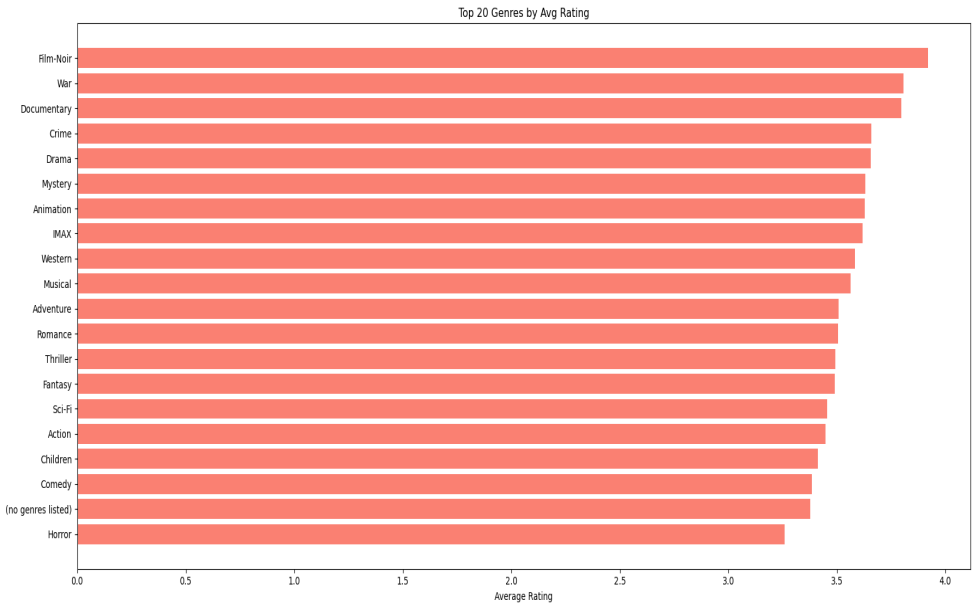
DATA EXPLORATION - VISUALIZATIONS

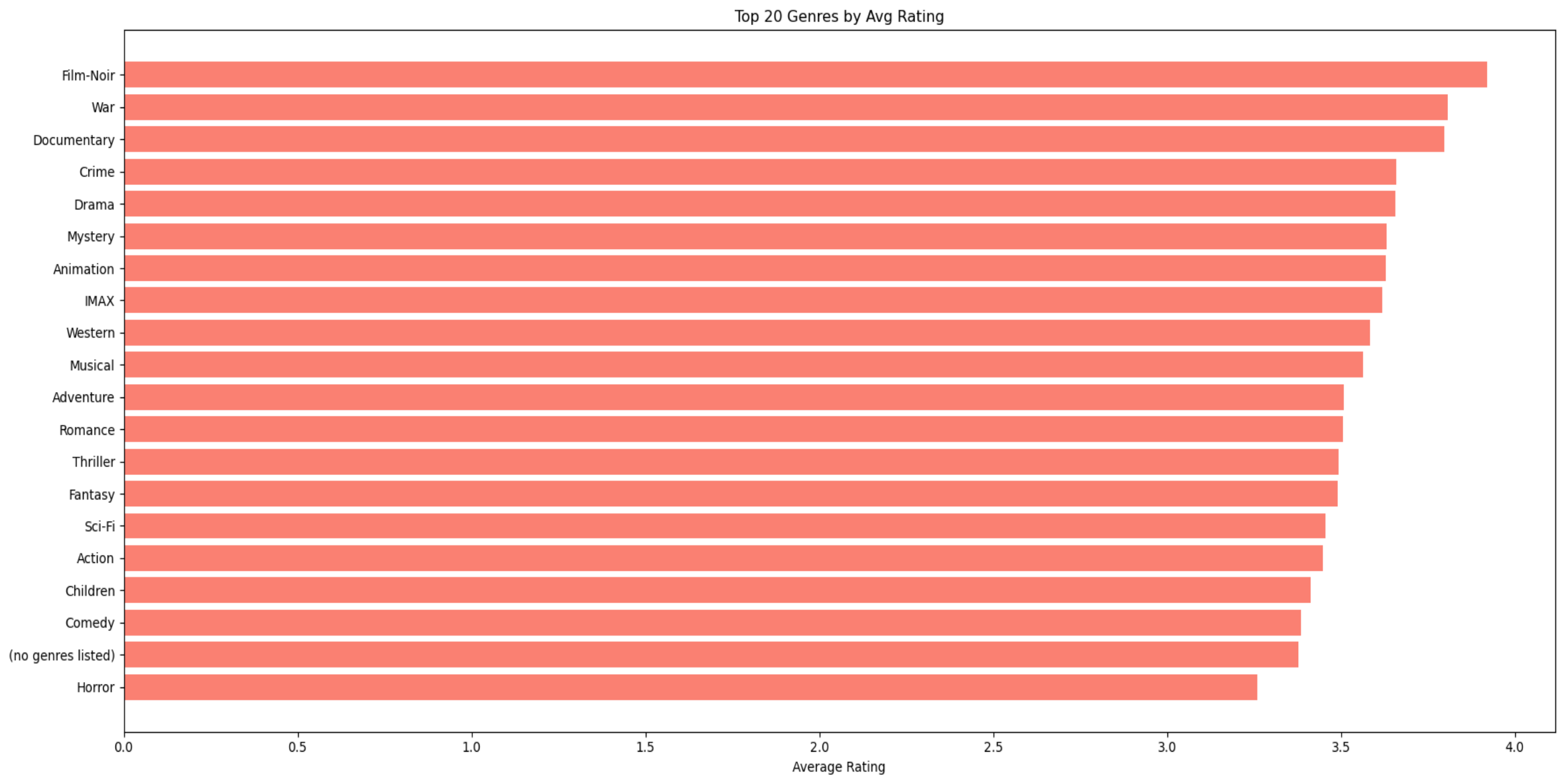
- We created visualizations to better understand the data.
- **Top 20 Movies by Avg Rating:**
- A bar chart showing movies like "Smokey and the Bandit" and "Star Wars" among the highest-rated, based on a small number of ratings.




Top 20 Movies by Avg Rating
My Left Eye Sees Ghosts (Ngo joh aan gin diy gwai)
Little Dieter Needs to Fly
Runaway Brain
Heidi Fleiss: Hollywood Madam
Tickling Giants
Vovka in the Kingdom of Far Far Away
A Detective Story
Travels of an Ant
Awfully Big Adventure, An
Cheburashka
Vacations in Prostokvashino
Obsession
Hollywood Shuffle
Fugitives
Priklyucheniya Kapitana Vrungelya
Going Places (Valseuses, Les)
Seems Like Old Times
Babes in Toyland
Match Factory Girl, The (Tulitikkutehtaan tyttö)
National Lampoon's Bag Boy
0
1
2
3
4
5
Average Rating

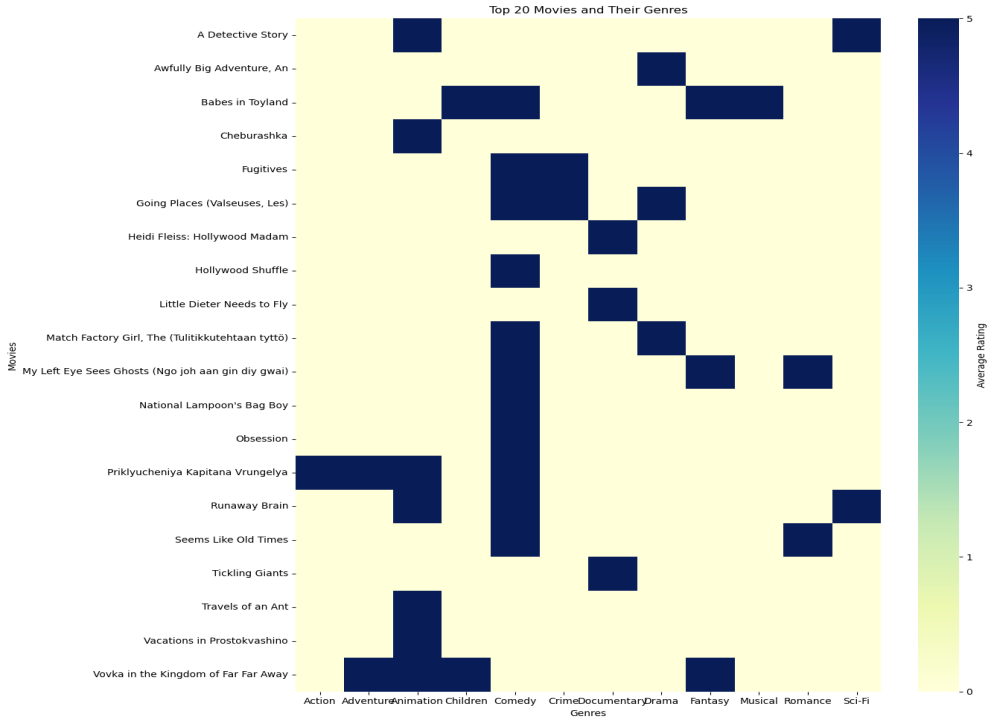
- **Top 20 Genres by Avg Rating:**
- A bar chart showing genres like "Film-Noir" and "War" have the highest average ratings, while genres like "Comedy" and "Adventure" have lower averages.

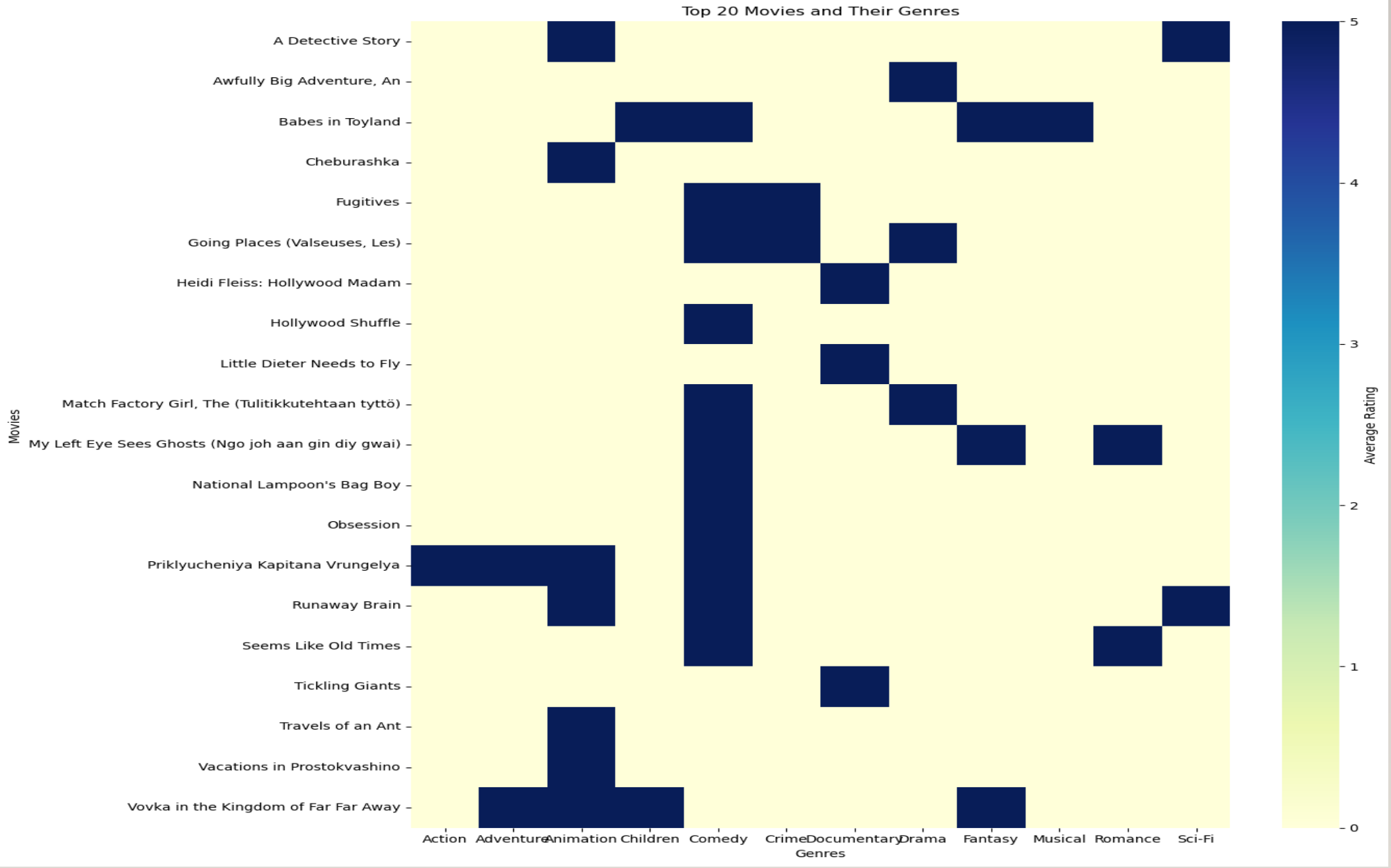



Top 20 Genres by Avg Rating
Film-Noir
War
Documentary
Crime
Drama
Mystery
Animation
IMAX
Western
Musical
Adventure
Romance
Thriller
Fantasy
Sci-Fi
Action
Children
Comedy
(no genres listed)
Horror
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Average Rating

GENRE HEATMAP:

•A heatmap visualizing the average ratings
for the top 20 movies across different genres.
•It helps identify which movies are strong across multiple genres (e.g., "Toy Story" excels
in Adventure, Animation, Children, Comedy, and Fantasy).




Top 20 Movies and Their Genres
A Detective Story
Awfully Big Adventure, An
Babes in Toyland
Cheburashka
Fugitives
Going Places (Valseuses, Les)
Heidi Fleiss: Hollywood Madam
Hollywood Shuffle
Little Dieter Needs to Fly
Match Factory Girl, The (Tulitikkutehtaan tyttö)
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National Lampoon's Bag Boy
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Movies
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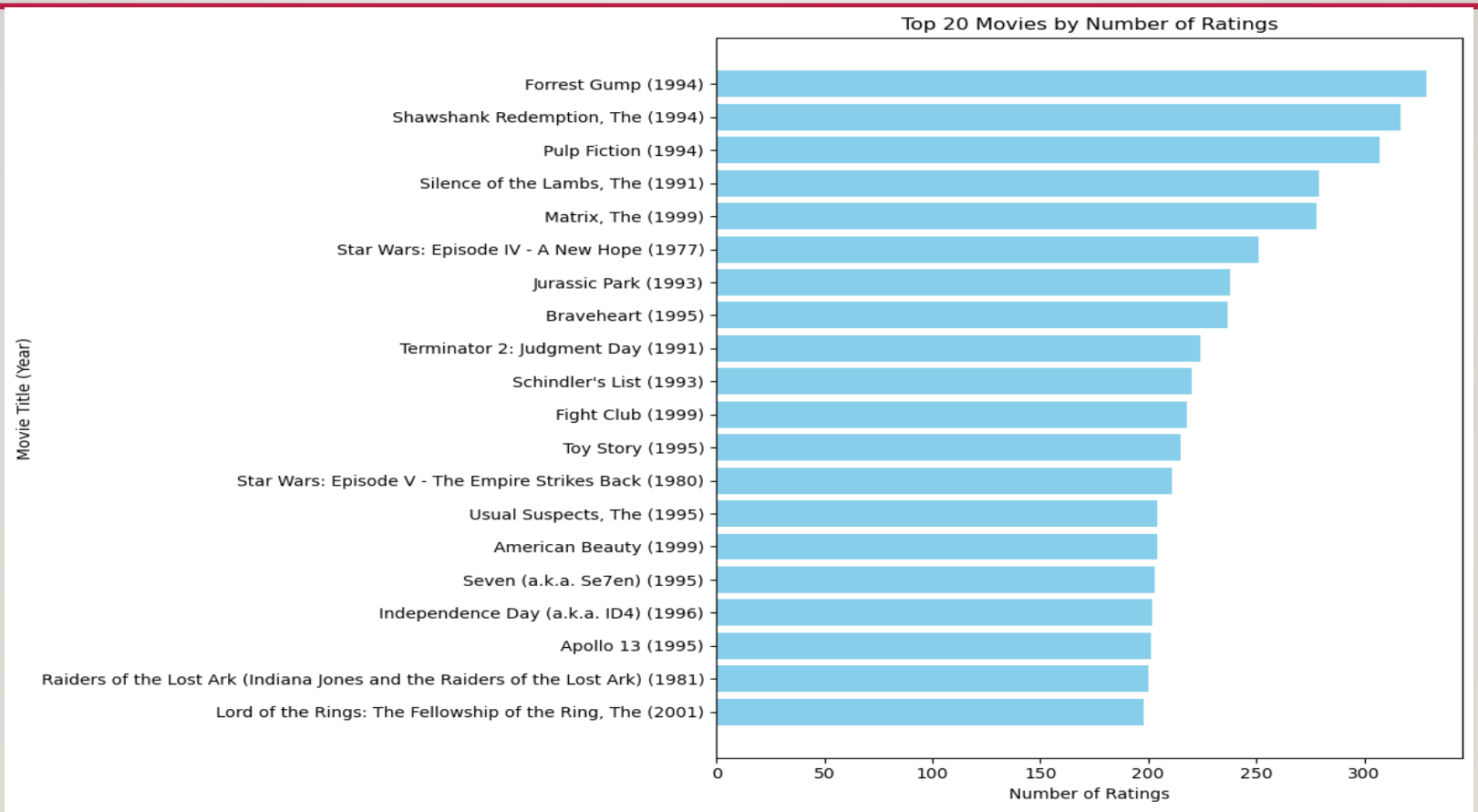
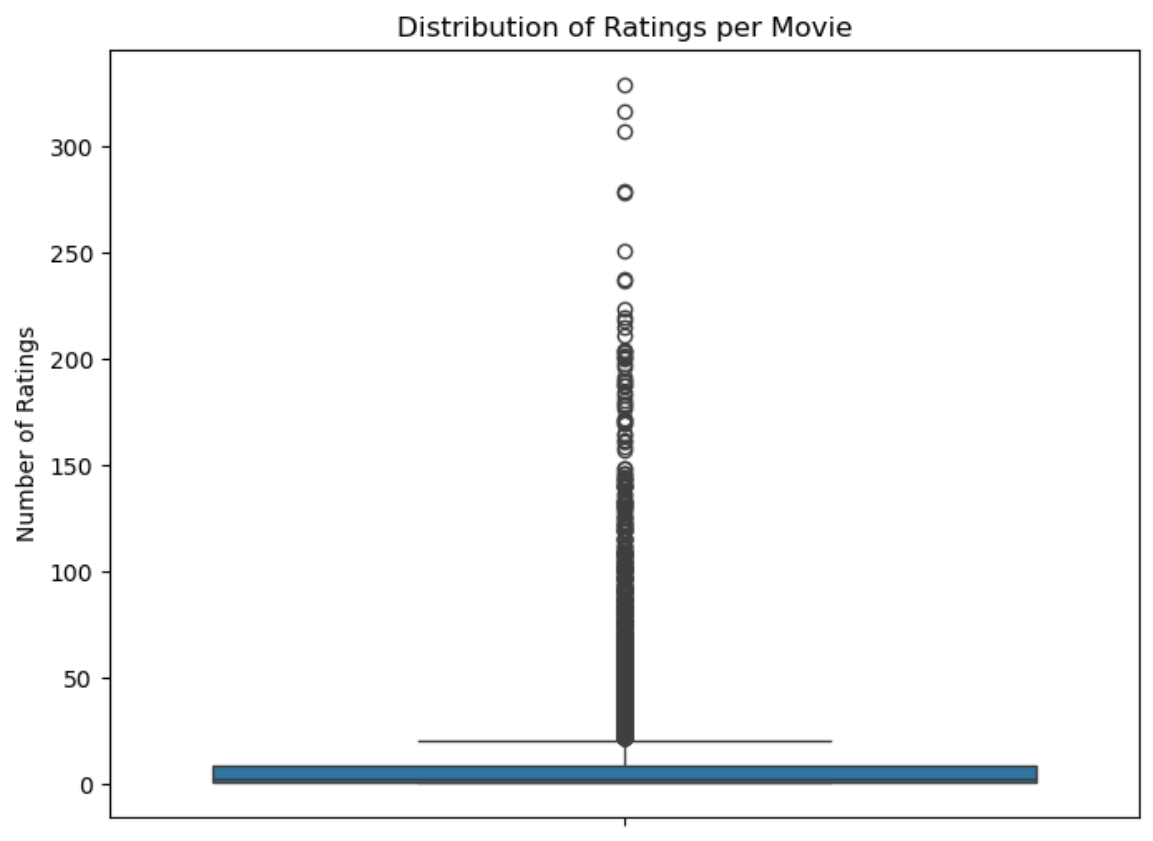
Children
Comedy
Crime
Documentary
Drama
Fantasy
Musical
Romance
Sci-Fi
Genres
Average Rating
0
1
2
3
4
5

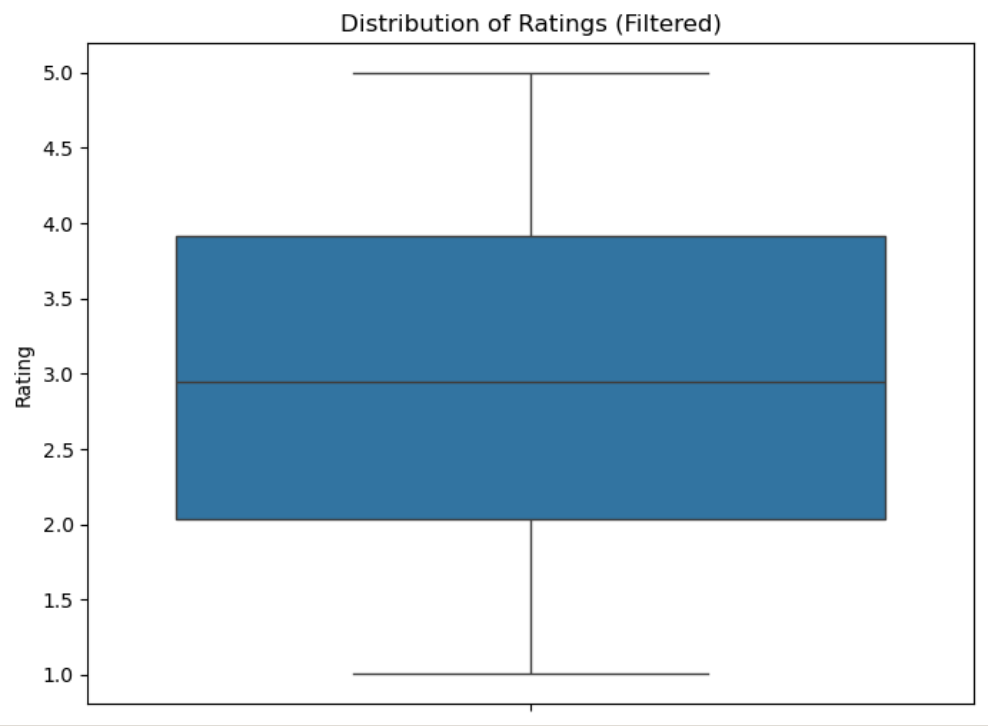
TOP 20 MOVIES BY NUMBER OF RATINGS - BARGRAPH



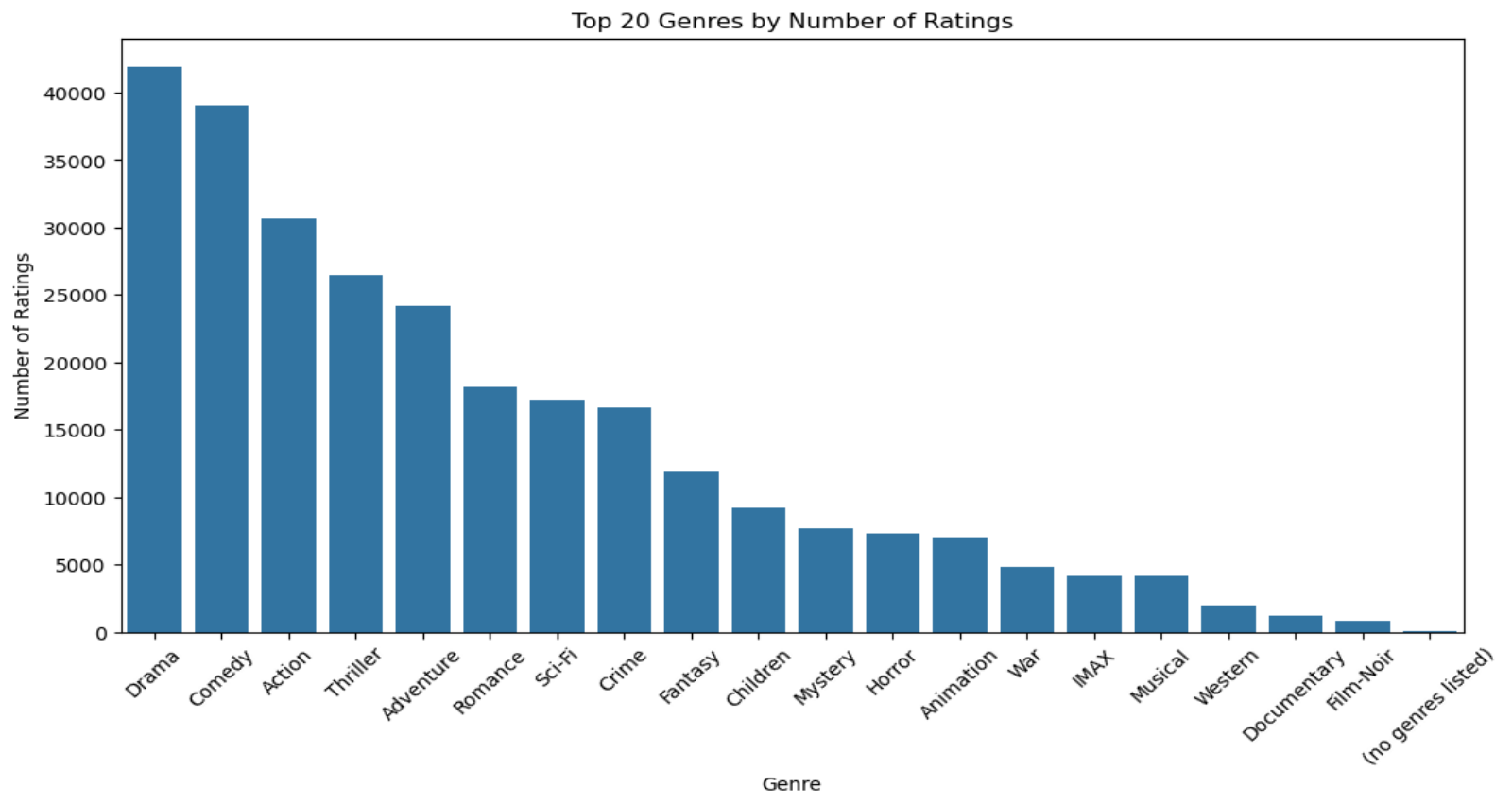
DISTRIBUTION OF RATINGS PER MOVIE - BOXPLOT



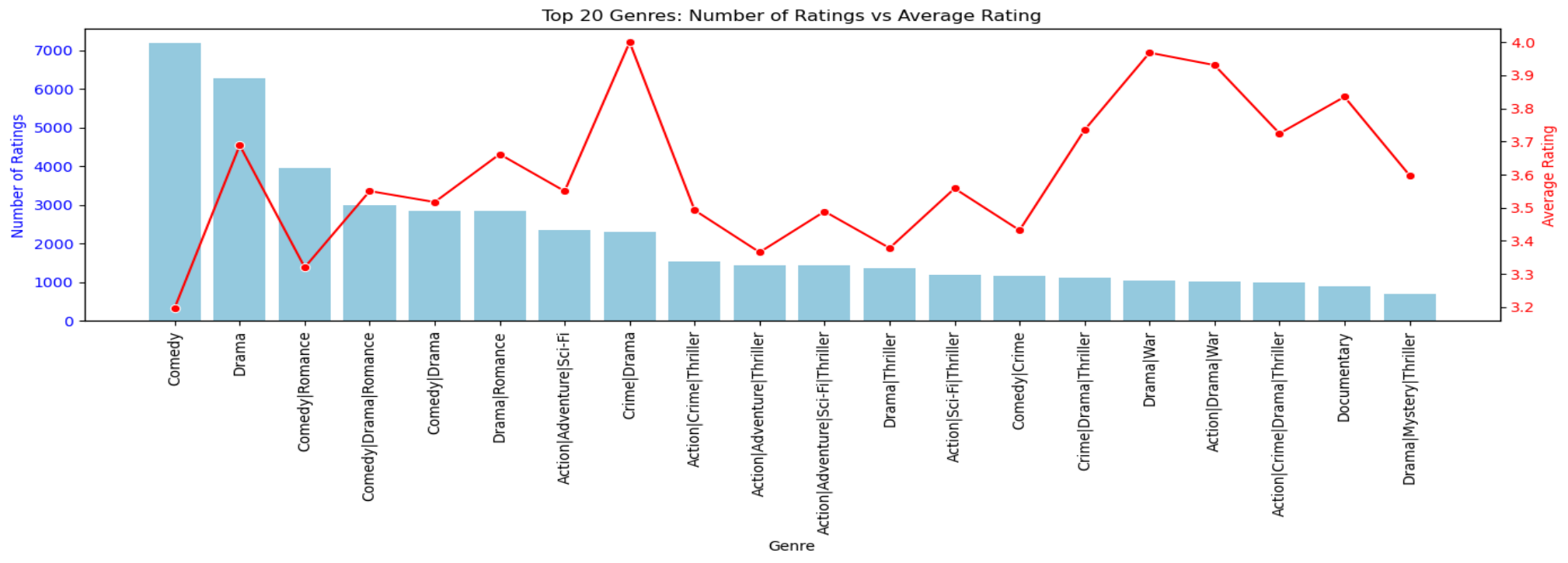
DISTRIBUTION OF RATINGS (FILTERED)



TOP 20 GENRES BY NUMBER OF RATINGS



TOP 20 GENRES: NUMBER OF RATINGS VS AVERAGE RATING



DATA EXPLORATION - RATING DISTRIBUTION

- We analyzed the distribution of ratings to understand rating behavior.
- **Ratings per Movie:**
- A boxplot shows that most movies have very few ratings, with a long tail of popular movies that have many ratings. The median number of ratings per movie is low.
- **Distribution of Ratings:**
- The histogram of ratings shows a roughly normal distribution, centered around 3.5 to 4.0.
- **Insight:** Users tend to rate movies they like, with a slight bias towards positive scores (3.5+).

DATA PREPROCESSING & FEATURE ENGINEERING

To prepare for modeling, we performed several preprocessing steps:

1. **Cleaned Titles:** Extracted the movie release year from the title into a separate Year column.

2. Converted Timestamps: The rating timestamp was converted to a readable rating year to analyze rating trends over time.

3. Handled Missing Data: Dropped 18 rows where the Year was missing from the title.

4. Feature Engineering for Content-Based Filtering:

•Created a new content column by concatenating the title and genres fields.

•This combined feature will be used to compute content similarity between movies (e.g., TF-IDF, Cosine Similarity).

MODELING APPROACH

We implemented and compared three different recommendation approaches:

1. Content-Based Filtering:

•**Logic:** Recommends movies similar to a specific movie based on its content (e.g., title and genres).

•**Method:** Used TF-IDF vectorization and cosine similarity on the content column.

•**Output:** For a given movie (e.g., Toy Story), it returns a list of movies with similar content.

CONT…

- **2. Collaborative Filtering (SVD):**
- **Logic:** Uses patterns from all users' rating history to predict what a user will like.
- **Method:** Applied Truncated SVD (50 components) to the user-item matrix to learn latent user and item features.
- **Output:** Predicts ratings for a user, even for movies they haven't seen.

CONT…

3. Collaborative Filtering (KNN):

•**Logic:** Recommends items similar to what a user has liked in the past.

•**Method:** Used NearestNeighbors on the transposed user-item matrix (movies as rows).

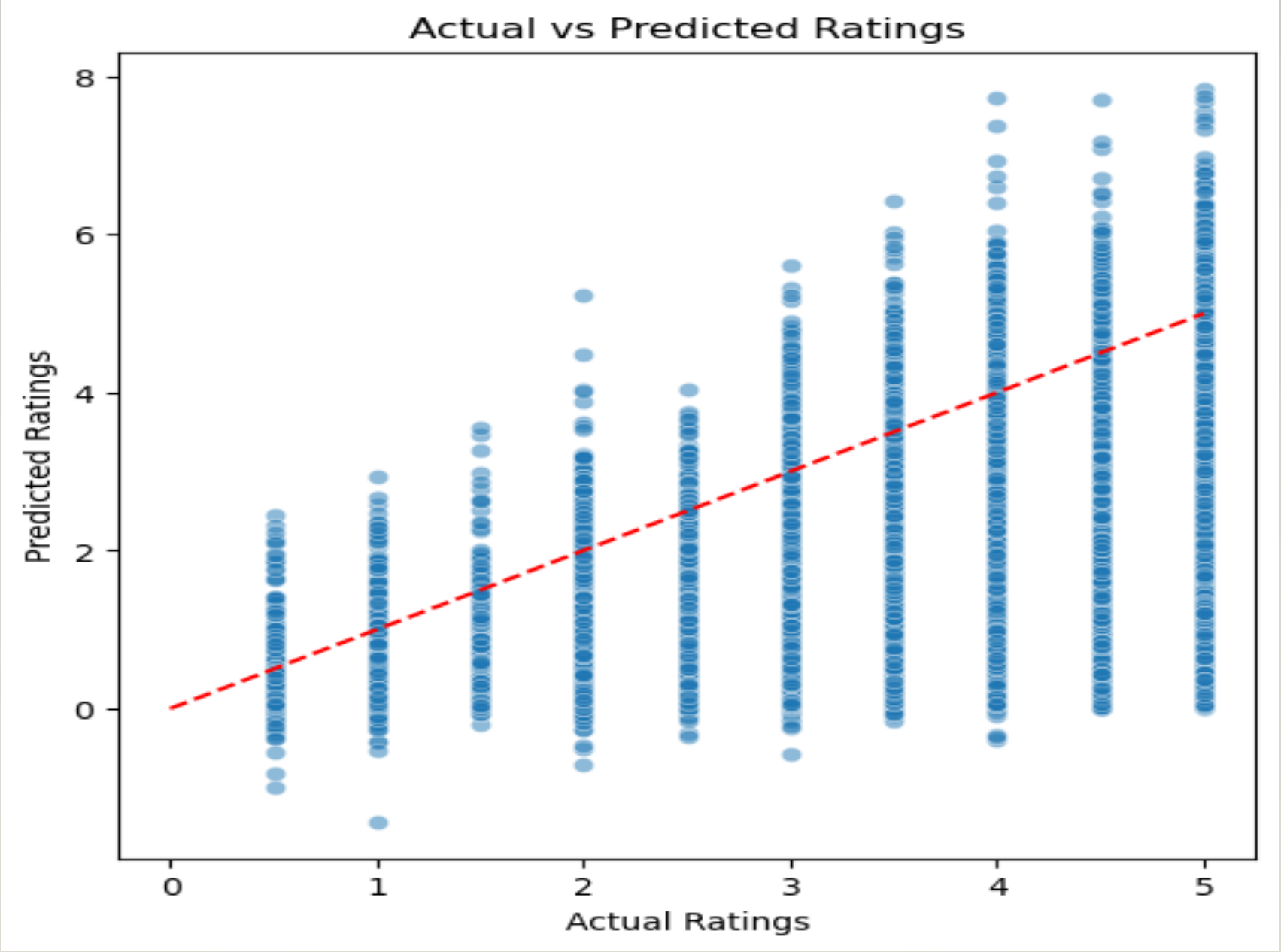
•**Output:** For a given movie, it finds similar movies; for a user, it aggregates recommendations
based on their highly rated movies.

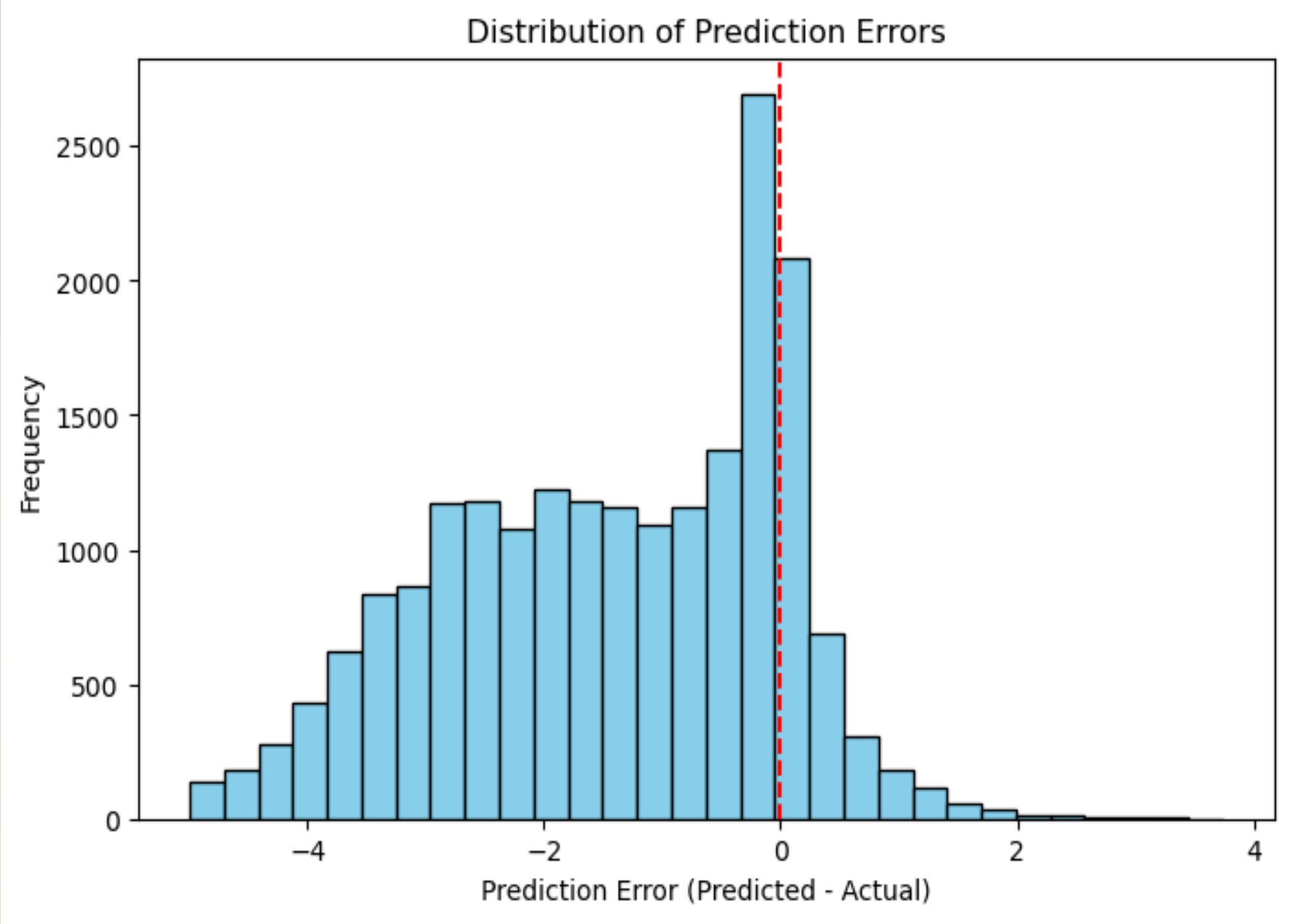
MODEL 1 - SVD EVALUATION

- We evaluated the SVD model's prediction accuracy.
- **Metrics:**
- **RMSE (Root Mean Squared Error):** 2.004
- **MAE (Mean Absolute Error):** 1.544
- **Interpretation:**
- On a 1-5 scale, the predictions are off by about 1.5 points on average (MAE).
- The RMSE is also relatively high, indicating some large errors in prediction.
- **Insight:** While absolute accuracy is moderate, the model can still capture relative patterns for ranking recommendations.

VISUALIZATIONS:

•**Scatter Plot (Actual vs. Predicted):** Shows a positive correlation but with significant spread.
•**Error Distribution:** The prediction errors are centered around zero but have a wide spread, confirming the moderate accuracy


Actual vs Predicted Ratings
Predicted Ratings
Actual Ratings
8
6
4
2
0
0
1
2
3
4
5


Distribution of Prediction Errors
Frequency
2500
2000
1500
1000
500
0
−4
−2
0
2
4
Prediction Error (Predicted - Actual)

MODEL 2 - KNN EVALUATION

- We implemented and tested an item-based KNN model.
- **How it works:**
- For each movie a user rated highly (≥ 3.5), we find similar movies using cosine similarity.
- Similarity scores are weighted by the user's original rating and summed.
- The top movies not yet seen by the user are recommended.

EXAMPLE OUTPUT FOR USER 1:

1. Ferris Bueller's Day Off (score: 58.02)
2. Terminator 2: Judgment Day (score: 36.39)
3. Aliens (score: 35.83)
4. Godfather, The (score: 33.69)
5. Die Hard (score: 31.40)

- **Interpretation:**
- The scores are not raw ratings but aggregated similarity scores. Higher scores indicate stronger recommendations.
- Recommendations are easy to explain: *"We recommend Die Hard because you liked similar action movies."*

HYBRID RECOMMENDER

We combined the strengths of SVD (latent factors) and KNN (explicit similarity) into a hybrid model.

Method:

•Scores from SVD and KNN are scaled (MinMaxScaler) and combined with weighted averaging.

•hybrid_score = svd_weight * svd_score + knn_weight * knn_score

Example Output for User 42:

1.Lock, Stock & Two Smoking Barrels (score: 0.85)

2.What Lies Beneath (score: 0.82)

3.Half Baked (score: 0.82)

4.Heat (score: 0.81)

5.Four Weddings and a Funeral (score: 0.81)

CONT…

- **Interpretation:**
- The hybrid model provides a balanced recommendation by blending global patterns (SVD) with local item similarities (KNN).
- The scores are normalized, so a score of 0.85 indicates a strong recommendation relative to the user's preferences

HYBRID MODEL EVALUATION

- We evaluated the hybrid model using standard recommendation metrics.
- **Metrics:**
- **Precision@5:** 0.0
- **Recall@5:** 0.0
- **F1@5:** 0
- **Interpretation:**
- For the specific user tested, none of the top 5 recommended movies were among the user's actual liked movies (ratings ≥ 4).

CONT…

- **Important:** This doesn't mean the model is broken. It could be that:
- The user's taste is very niche or difficult to predict.
- The user has already rated all the movies the model would recommend.
- The evaluation metric (precision/recall) is very strict, and the user may still enjoy the recommended movies even if they haven't rated them.

CONT…

Next Steps:

•Evaluate on a larger set of users to get average metrics.

•Tune the svd_weight and knn_weight parameters to find the optimal blend.

•Consider using other metrics like NDCG (Normalized Discounted Cumulative Gain) which accounts for ranking order.

CONCLUSION & BUSINESS IMPACT

- **Key Findings:**
- The data shows a clear opportunity for personalization. Users have diverse tastes and rating patterns.
- Content-based filtering is useful for "similar to this movie" features.
- Collaborative filtering (SVD and KNN) provides powerful user-personalized recommendations.
- A hybrid approach combines the strengths of both, offering more robust recommendations.

BUSINESS IMPACT FOR JACARANDA MOVIE SHOP:

•**Increased Engagement:** By showing customers movies they are likely to enjoy, they will watch longer and more often.
•**Improved Satisfaction:** Personalized recommendations make users feel understood, increasing loyalty.
•**Reduced Churn:** A "sticky" platform that consistently provides value will retain more customers.
•**Revenue Growth:** More engagement and satisfaction can translate to higher purchase rates, subscription renewals, and cross-selling opportunities.

RECOMMENDATION:

- Implement the hybrid recommender in production.
- Conduct A/B testing to measure the impact on user engagement and business metrics.
- Continuously iterate and improve the model with new data and user feedback.